



Rotate each shape. Answer with the new coordinates.

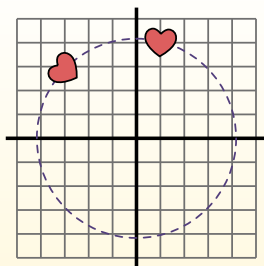
θ = Angle of Rotation

Rotation Formula

$$x1 = x \cos(\theta) - y \sin(\theta)$$

$$y1 = x \sin(\theta) + y \cos(\theta)$$

In the example to the right the shape is at coordinates (1,4). Lets find the coordinates if we rotated the shape 60°.



$$1. \quad x1 = 1 \times \cos(60) - 4 \times \sin(60)$$

$$y1 = 1 \times \sin(60) + 4 \times \cos(60)$$

$$2. \quad x1 = 1 \times 0.5 - 4 \times 0.87$$

$$y1 = 1 \times 0.87 + 4 \times 0.5$$

$$3. \quad x1 = 0.5 - 3.48$$

$$y1 = 0.87 + 2$$

$$4. \quad x1 = -2.98$$

$$y1 = 2.87$$

5. Looking at shape, we can see that rotated 60° it is at (-2.98, 2.87).

Answers

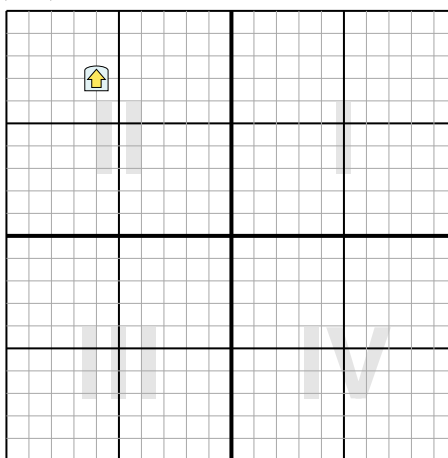
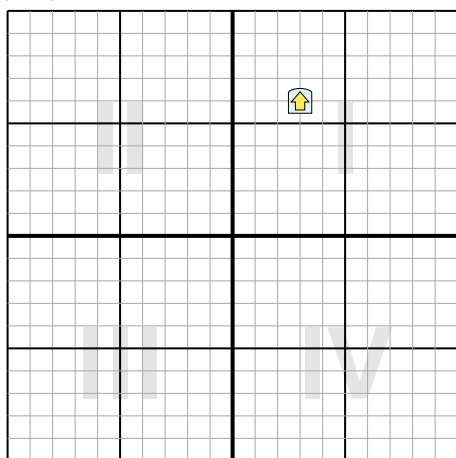
1. _____

2. _____

3. _____

4. _____

- 1) Rotate the shape 231° around the point (0,0). 2) Rotate the shape -205° around the point (0,0).



- 3) Rotate the shape -134° around the point (0,0). 4) Rotate the shape -224° around the point (0,0).

